

Optimizing Credit Allocation and Stress-Testing Bank Exposure Under Semicon 2.0:

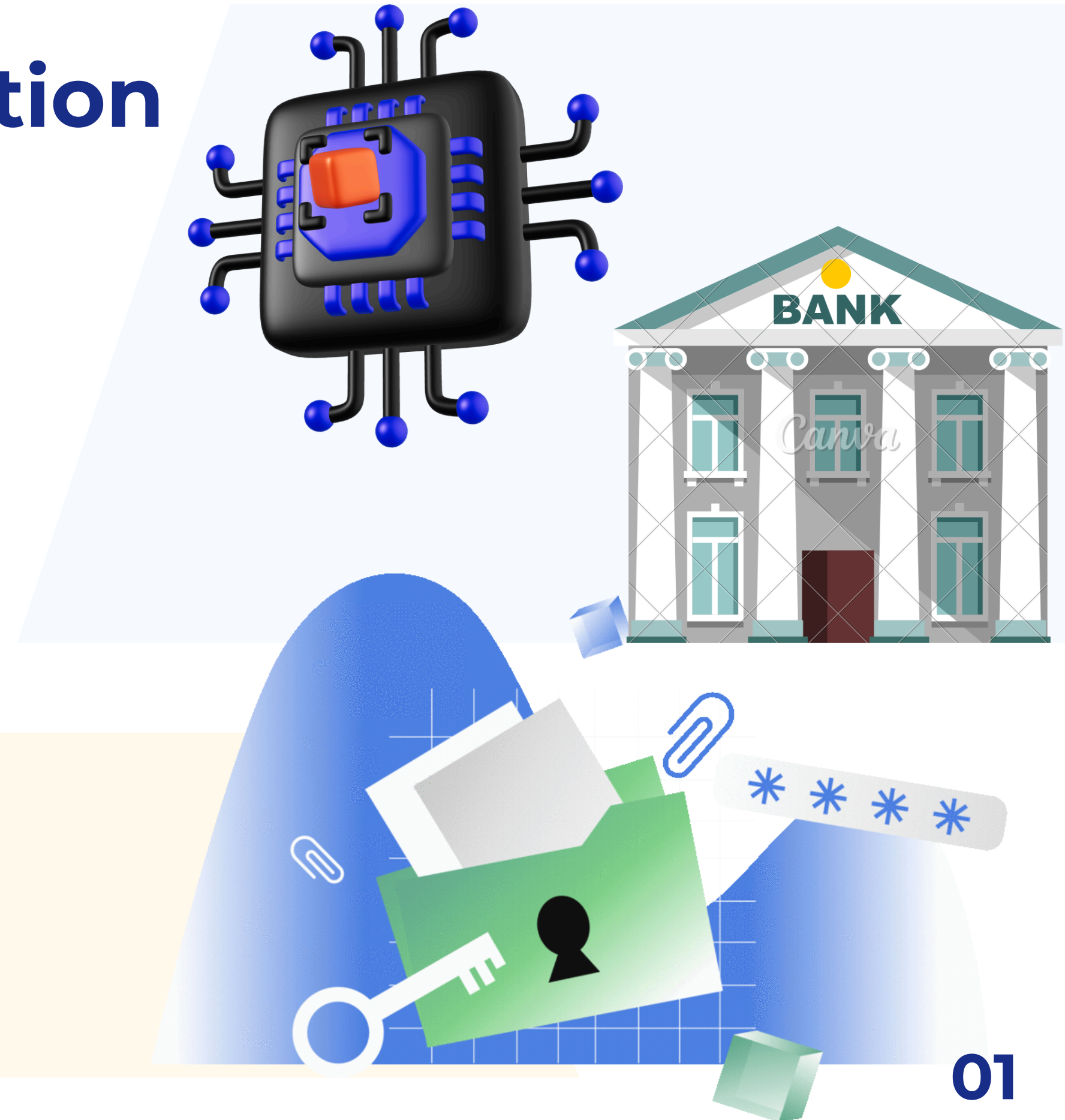
A Machine Learning Framework for
Macro-Prudential Risk Management

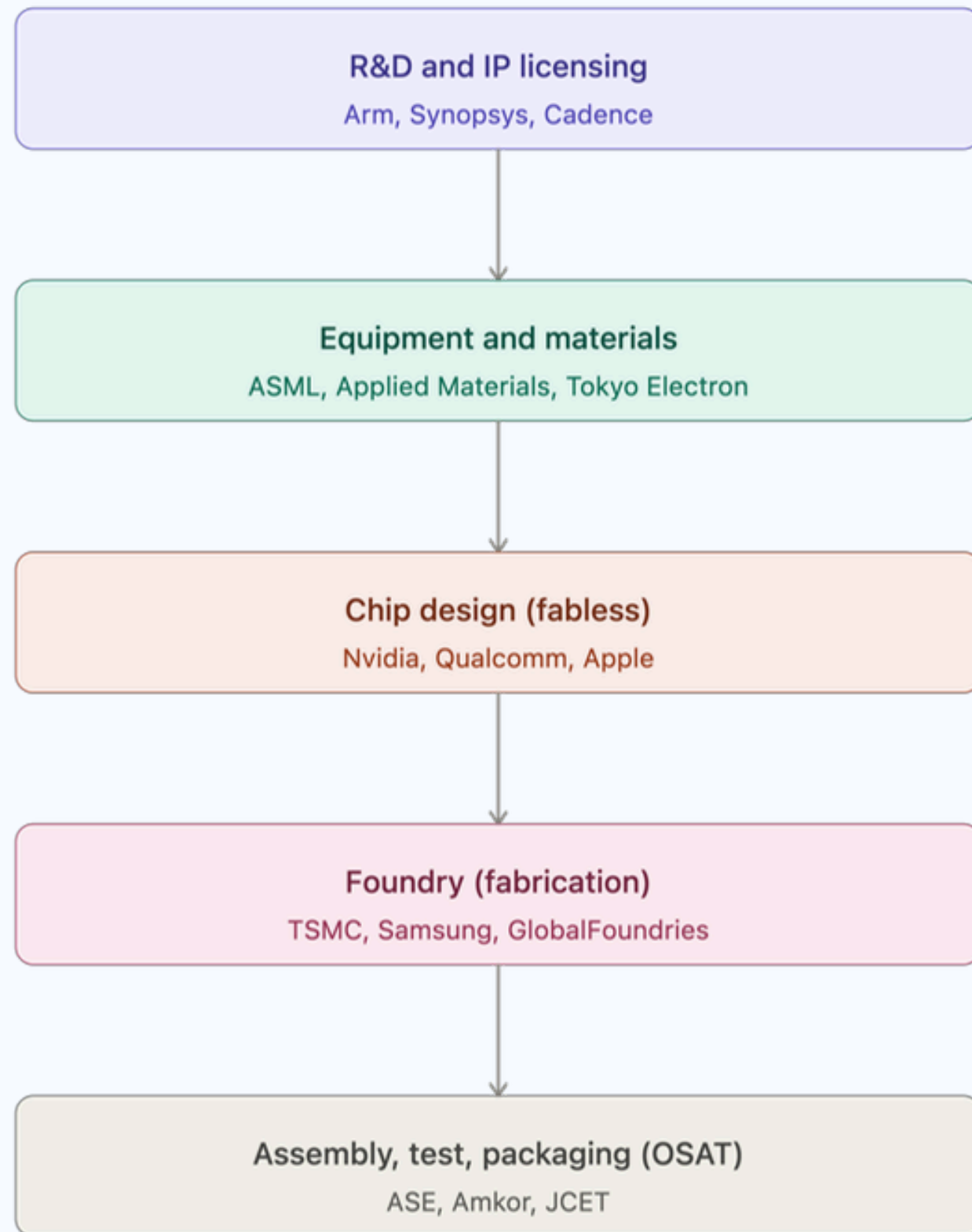
Presented By :

G.Shashank Yadav
Hansitha Purshotham
A Rahul
G Aishwaryambica

Mentor:

Prof Geetha Tripathi





Global Semi-Con Value

Why it matters

Semiconductors underpin nearly every modern technology computing, telecom, defense, autos, AI -making the value chain a foundation of both economic competitiveness and national security.

Chain

Strategic Reach

The chain is highly concentrated at critical nodes (ASML for lithography, TSMC for leading-edge fabrication, Arm/Synopsys/Cadence for design IP and tools), so a disruption at any single point — geopolitical, natural disaster, or export controls — can stall global chip supply.

Demand Drivers

Growth is being pulled by AI and data center buildout, 5G/connectivity, EV and automotive electrification, and continued smartphone/consumer electronics upgrade cycles.

What is Semicon 2.0?

Semicon 2.0 is the Government of India's next-generation initiative to strengthen the country's semiconductor ecosystem. It aims to promote chip design, semiconductor manufacturing, packaging, research & development, and skill development, while attracting global investments and reducing import dependency.



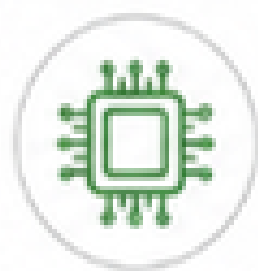
₹1.27 LAKH CRORE

Proposed Investment to Strengthen India's Semiconductor Ecosystem

Key Focus Areas



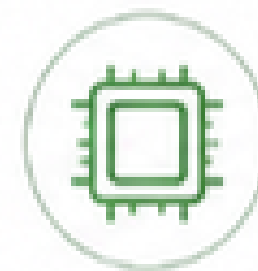
**SEMICONDUCTOR
MANUFACTURING**



**CHIP DESIGN
& INNOVATION**



**RESEARCH &
DEVELOPMENT**



**ASSEMBLY, TESTING &
PACKAGING (ATMP)**



**SKILL DEVELOPMENT
& TALENT CREATION**



**INVESTMENTS &
PARTNERSHIPS**



Problem Statement

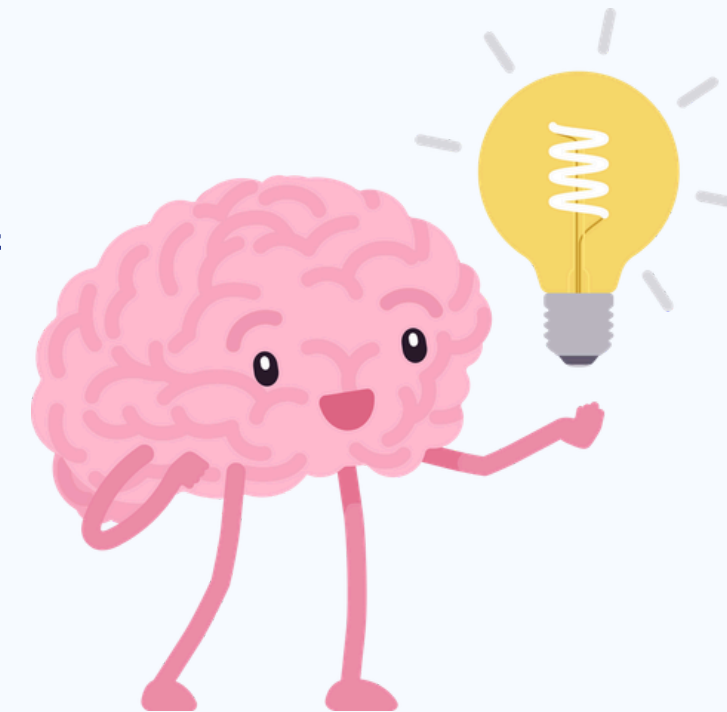
Current Challenges:

Traditional banking systems face several limitations while financing semiconductor projects. These include:

- Manual credit evaluation
- High uncertainty in technology projects
- Lack of intelligent risk prediction
- Difficulty in assessing startup innovation potential
- No integrated macroeconomic stress testing
- Limited transparency in credit allocation

Where do we Stand at

There is currently no integrated AI-driven framework capable of intelligently evaluating semiconductor financing proposals while simultaneously assessing bank exposure under different macroeconomic conditions.



Project Objectives

Our Primary Objective:

01 Develop a Machine Learning based decision-support framework that assists banks in financing semiconductor projects under India's Semiconductor Mission 2.0.

Other Objectives:

02

Study semiconductor financing ecosystem.

03

Collect financial and macroeconomic datasets.

04

Develop intelligent credit scoring models.

05

Predict project risk using Machine Learning.

06

Perform macroeconomic stress testing.

07

Recommend optimized credit allocation.

Literature Review

Existing Research

Semiconductor Industry Research focuses on

- Manufacturing
- Supply chain
- Government incentives
- Limited work on financing decisions.



Banking

Traditional credit scoring relies on

- Financial ratios
- Historical repayment
- Expert judgement

Limited support for innovation-based industries.

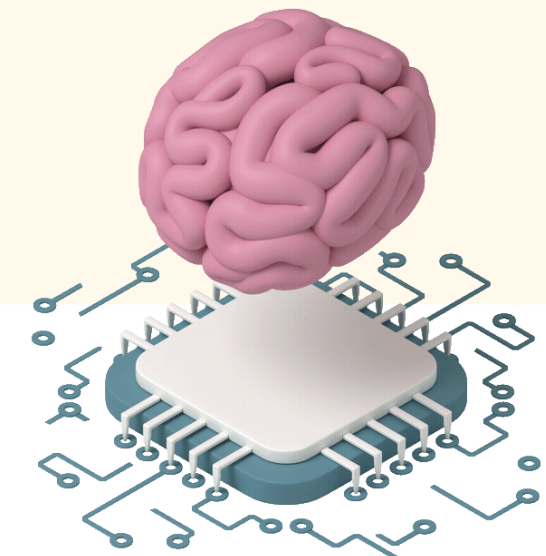


Research Gap:

Current literature lacks an integrated framework that combines:

- Semiconductor financing
- Credit allocation
- Machine Learning
- Stress testing
- Macroeconomic analysis

Our project addresses this gap.



What Makes This Approach Novel

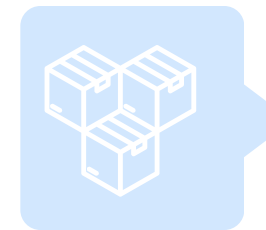
Traditional credit assessment models rely mainly on financial indicators and overlook the unique risks of the semiconductor industry. Our framework bridges this gap using machine learning and sector-specific risk factors.

Sector-Specific ML



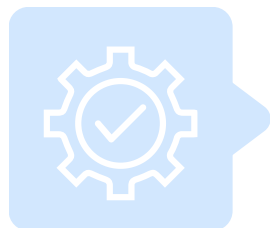
Designed exclusively for semiconductor financing, not a generic credit model

Multi-Dimensional Inputs



Financial, technological, and innovation factors evaluated simultaneously

Milestone-Based Lending



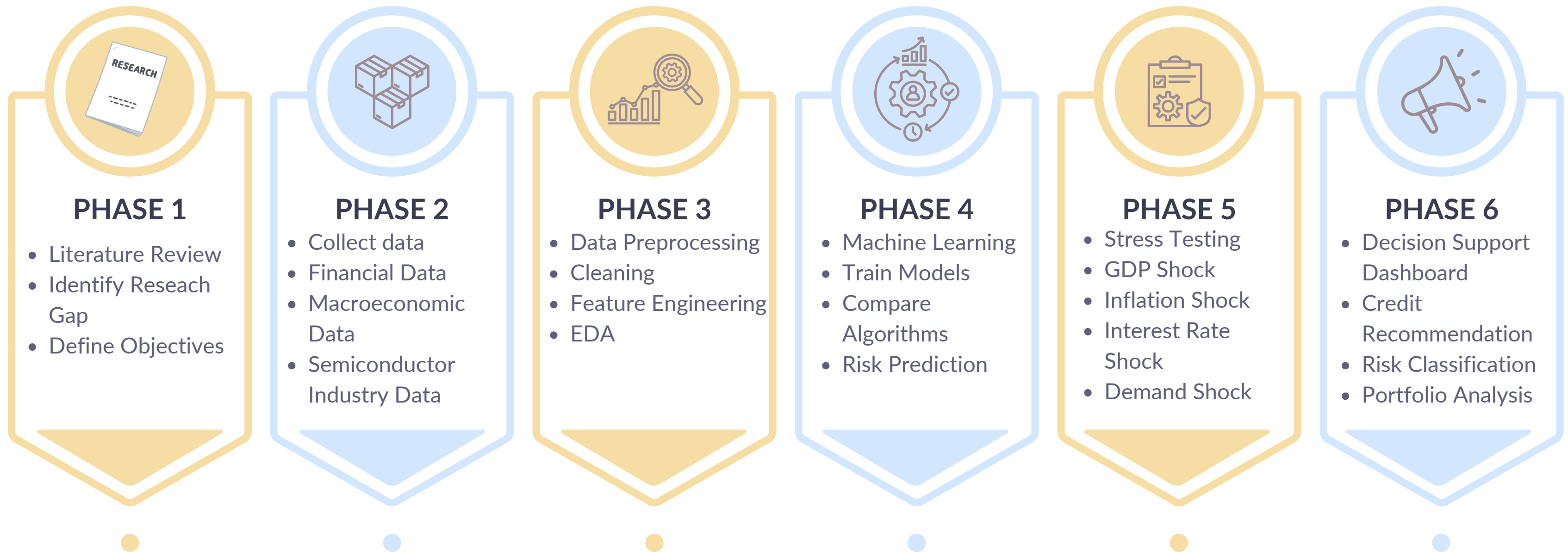
Disburses funds against R&D and production milestones, not one-time approval.

Macro-Prudential Testing

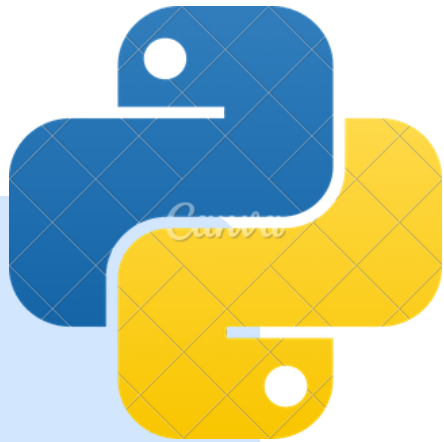


Estimates bank exposure across economic scenarios before capital is committed

Methodology



Tools & Technologies



Category	Technology
Programming	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn, XGBoost
Visualization	Matplotlib, Plotly
Dashboard	Streamlit
Database	PostgreSQL / CSV
Version Control	Git & GitHub
Cloud	Google Cloud (Future Scope)



Project TimeLine

The project will be executed in seven structured phases, ensuring a systematic approach from research to final implementation.



STEP 1

Literature Review

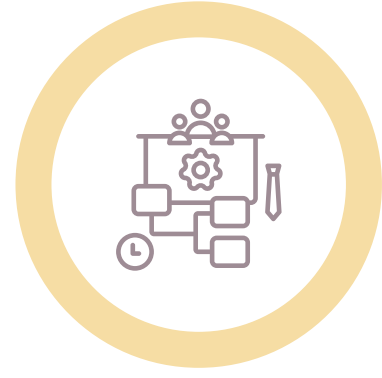
Review literature, understand Semicon 2.0, identify research gap and define project scope.



STEP 2

Data Collection

Gather financial, technological, and macroeconomic data required for the project.



STEP 3

Data Preparation

Clean, preprocess, and engineer features for machine learning.



STEP 4

Model Development

Build and train the ML-based credit scoring model.



STEP 5

Risk Analysis Framework

Perform stress testing and optimize credit allocation strategies.



STEP 6

Prototype Implementation

Develop the prototype/dashboard and evaluate model performance.



STEP 7

Final Documentation

Prepare the report, finalize results, and present the project.

Current Progress

Current progress focuses on understanding the semiconductor financing ecosystem, identifying research gaps, and preparing the proposed machine learning framework.



Individual Responsibilities and Contributions

Team Member	Responsibility
Hansitha Purshotham	Literature Review, Problem Statement, Research Gap, Documentation
A.Rahul	Dataset Collection, Data Preprocessing, Exploratory Data Analysis
G.Shashank Yadav	Machine Learning Model Development, Model Evaluation, Explainability
G Aishwaryambica	Stress Testing Framework, Dashboard Development, Presentation, GitHub Repository

Conclusion and Future Score

Key Takeaway

- Improves credit decisions using Machine Learning.
- Enhances bank risk assessment through stress testing.
- Supports smarter financing for Semicon 2.0.

Future Scope

- Integrate real-time financial and market data feeds
- Expand framework to other high-tech industries
- Develop a live dashboard for bank deployment
- Enhance accuracy with Deep Learning and Explainable AI

Thank You For Your Valuable Time.

We welcome your questions and feedback.

Presented By :

Shashank Yadav
Hansitha Purshotham
A Rahul
G Aishwaryambica

